Theory and Methodology of Science

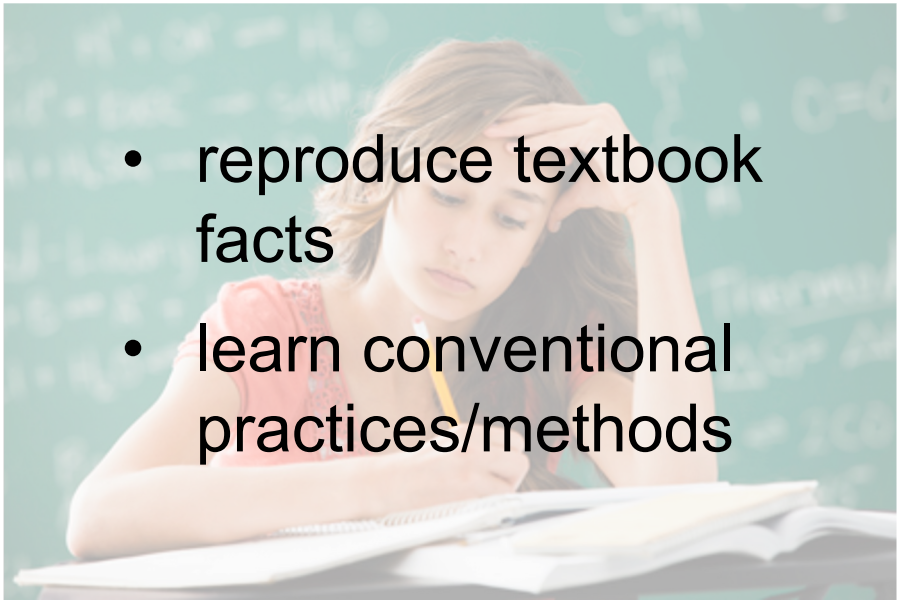
Introduction Lecture

Till Grüne-Yanoff



An Important Transition

Student → Scientist/ Professional



- reproduce textbook facts
- learn conventional practices/methods



- choose best method for your goals
- justify method choice

Method Choice – Examples from MSc Theses



Your task is to identify improvements in the energy performance of a type of buildings. Should you do this by…

FE simulation, or …
a case study, or…?



You want to investigate wear of a new road cover. Should you do this by…

laboratory experiment, or..
field experiment, or…?

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science



Methods of science

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science



Methods of science

FE simulation X-ray spectroscopy Questionnaire
Temperature measurement Pen-and-paper-models
Significance testing Optimization under constraints
Process tracing Regression analysis

Different Levels of Discussing Method Choice



General philosophy of science

- How is it possible to acquire knowledge of the world?
- Are our best theories true, or are they merely useful?
- Everything we observe is interpreted on the basis of our beliefs and expectations; how is it possible to overcome this and be *objective*?

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science

- Which of the available methods shall I choose for my goals?
- What justifies this choice of method?

The Focus of TaMoS



Methodology: investigates assessment and justification of *method choice*

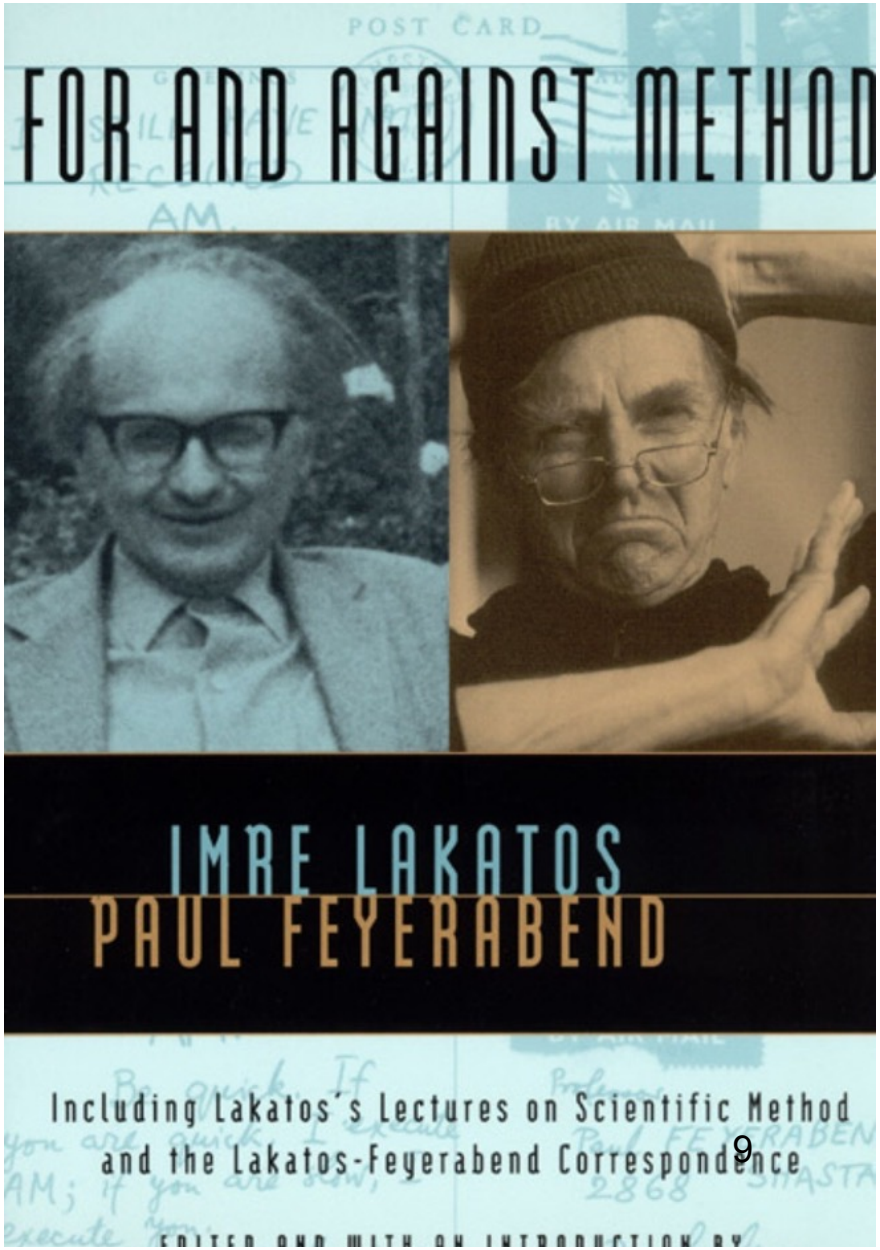
Methodology ≠ Collection of methods



Methods of science

Why *Justify* One's Method Choice?





Why *Justify* One's Method Choice?

- **Self-reflect** – ensure that you think your choice is the best
- **Plan** – convince your future selves that they should implement your choice
- **Coordinate** – motivate others to join in on your approach
- **Persuade** – give others reasons to trust your results
- **Be accountable** – document your motivations for making this choice
- **Epistemically productive** – there is no knowledge without justification

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements:
  - Prepare for and attend seminars
  - Pass exam
  - (Fulfil project requirements)

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- Credit requirements:
  - **Prepare for and attend seminars**
  - Pass exam
  - (Fulfil project requirements)

- 4 mandatory seminars
- One per week, starting 9/9
- Many seminar groups offered each week
- Register on CANVAS
- Hand in prep quiz before seminar

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements:
  - Prepare for and attend seminars
  - **Pass exam**
  - (Fulfil project requirements)

- Open book pen & paper exam
- No computer, no online access
- Three parts:
  - Conceptual understanding (MCQs)
  - Short essay questions
  - Longer essay questions (optional, only if you meet threshold for 1&2)

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements:
  - Prepare for and attend seminars
  - Pass exam
  - **(Fulfil project requirements)**

- For example:
  - mixed individual and group assignment
  - popularizing summary & methodological analysis of scientific article from your field

The Structure of TaMoS

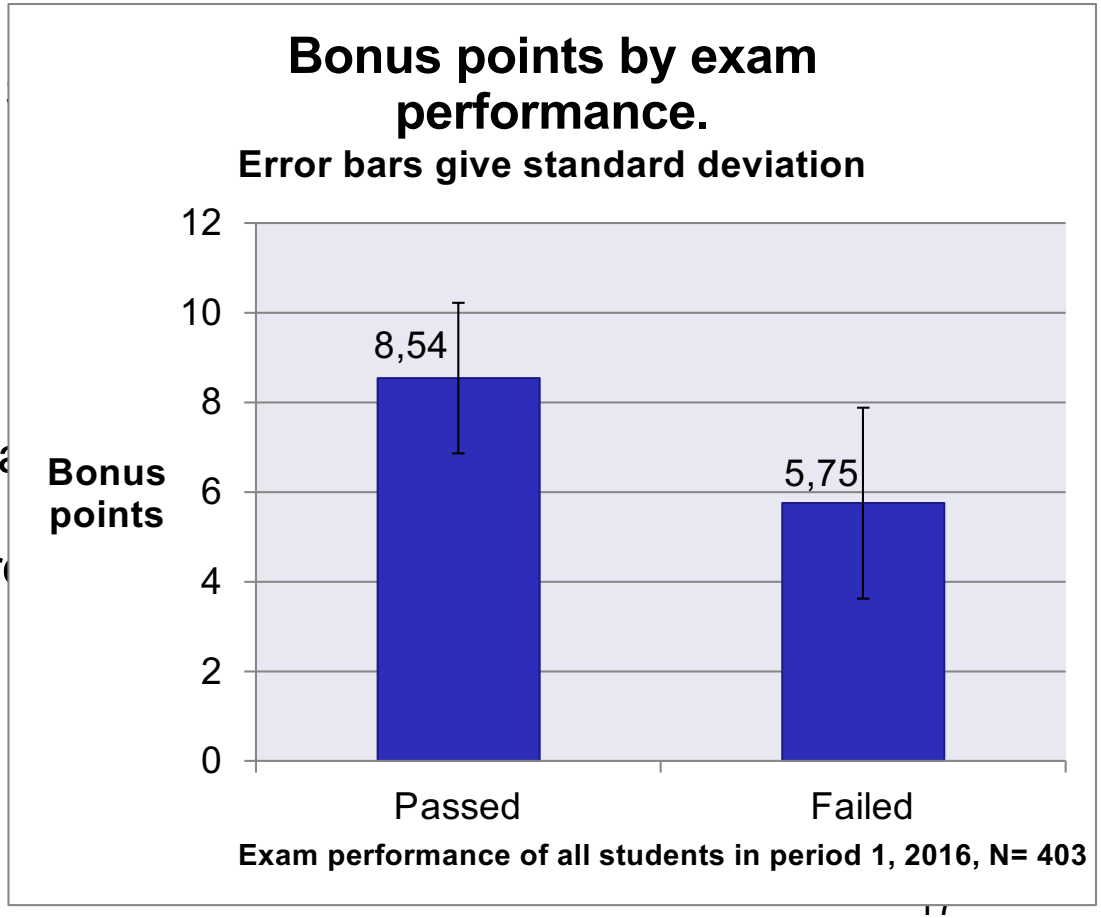
- Different course codes – different structure and requirements
- Credit requirements – seminars, exam, (project)
- Additional offers:
  - Introduction lecture
  - 10 Video lectures (with ca. 5 chapters each) – bonus points for quiz answers
  - 2 Flipped classrooms – bonus points for participation
  - Script covering all video lectures & additional reading material

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements – seminars, exam, (project)
- Additional offers – campus/video lectures, bonus points
- Bonus points
  - Max 5 points
  - Counts towards part 1 of the exam
  - Good for this term’s exam and re-exam

The Structure of TaMoS

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- Bonus points
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The Structure of TaMoS

- All information on CANVAS – make sure you have access
  - Schedule
  - Videos
  - Quizz questions
  - Script
  - Old exams
  - Teacher contact

BREAK

Different Ways of Justifying Method Choice

1. *The* scientific method: "I choose the method that is the scientific one – in contrast to non- or pseudoscientific ones"

2. Conventions: "I choose the methods my teachers / my peers choose"

3. Epistemic Tools: "I choose the method that tends to best satisfy my epistemic goals"

Different Ways of Justifying Method Choice

1. *The* scientific method: “I choose the method that is the scientific one“

Problem 1: Diversity of science

- Different goals
- Different methods
- Different standards

Different Ways of Justifying Method Choice

1. *The* scientific method: “I choose the method that is the scientific one“

Problem 2: Lack of an accurate demarcation

- E.g. Popper’s falsifiability criterion

Different Ways of Justifying Method Choice

2. Conventions: “I choose the methods my teachers / my peers choose”

Some problems:



How to correct mistaken theories or methods that are dominant in your field?



How to collaborate with other disciplines that have different conventions?

Different Ways of Justifying Method Choice

3. Epistemic Tools: “I choose the method that tends to best satisfy my epistemic goals”

- Take scientists' actual epistemic goals
- Conceptual analysis: what properties important for which goals
- Instrumental rationality: choose methods that best satisfy these properties!
- Which methods are best instruments? Conceptual + empirical question

Different Ways of Justifying Method Choice

3. Epistemic Tools: "I choose the method that tends to best satisfy my epistemic goals"

- Reasons: considerations that justify choosing a method (over others) for a given goal

Method A	
pro	*con*
…	…

Method B	
pro	*con*
…	…

- Choose the method that has the stronger supporting or the weaker contradicting reasons

Epistemic Tools Justification - Illustration



Your task is to investigate wear of a new road cover.

Should you do this by…

- laboratory experiment, or field experiment, or…?

Goal: *explain* why the new cover wears differently than the previous one in Sweden

Analysis: explanation requires identifying the difference-making causes

Comparison:

Lab experiment	Field experiment	Observational study
Good tool to identify causes	Good tool to identify causes	Less good tool to identify causes
Good background variable control	Less good background variable control	Less good background variable control
Difficult to generate actual background conditions	Set in actual background conditions	Set in actual background conditions

What are the Goals of Science?

Predicting X	Explaining X	Designing X
Knowing that at time t, X will occur	Knowing the cause(s) that produced X	Knowing that artifact X will satisfy functions F

Scientific Knowledge

Classical Definition of Knowledge

Knowledge is true, justified belief

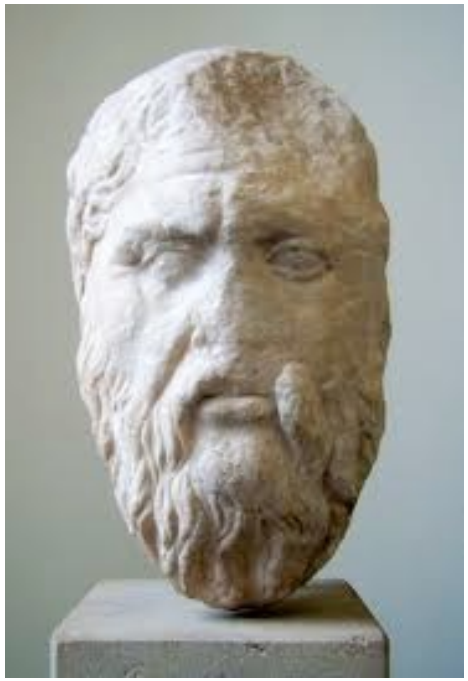
A knows that *p* only if

i. *A* believes that *p*.

ii. *A* is justified in believing that *p*.

iii. *p* is true.

(Not sufficient! Look up *Gettier problems*)



Plato (427-347 BC)

Classical Definition of Knowledge

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- Information that no-one believes to be true cannot be knowledge
- All knowledge is held by agents who act based on their beliefs

Classical Definition of Knowledge

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A knows that *p* only if

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ii. *A* is justified in believing that *p*.

iii. *p* is true.

- You might accidentally believe the truth – that is *not* knowledge
- Knowing requires providing valid reasons for your beliefs
- That includes providing valid reasons for having chosen the methods that provide evidence

Classical Definition of Knowledge

Knowledge is true, justified belief

A knows that *p* only if

i. *A* believes that *p*.

ii. *A* is justified in believing that *p*.

iii. *p* is true.

If the claim *p* is false, then it cannot be knowledge…

…even if you believe it to be true

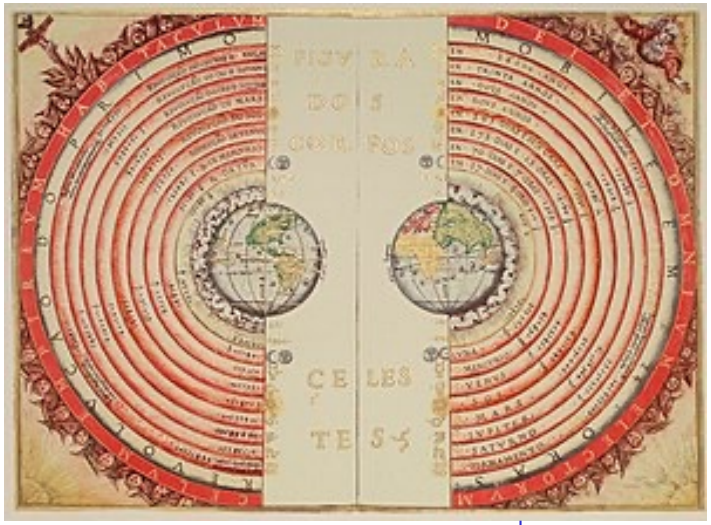
…even if your belief in its truth is justified

Classical Definition of Knowledge

If a belief is justified, it does *not* necessarily follow that this belief is true

Tycho Brahe
1546-1601





Geocentric model of the universe by B. Velho, 1568

Example:

Astronomy before the telescope

Best available sensory experience did not suffice to justify rejection of Geocentric model



Galileo's telescope, c. 1610

Uncertainty of Knowledge

Knowledge can be and often is uncertain

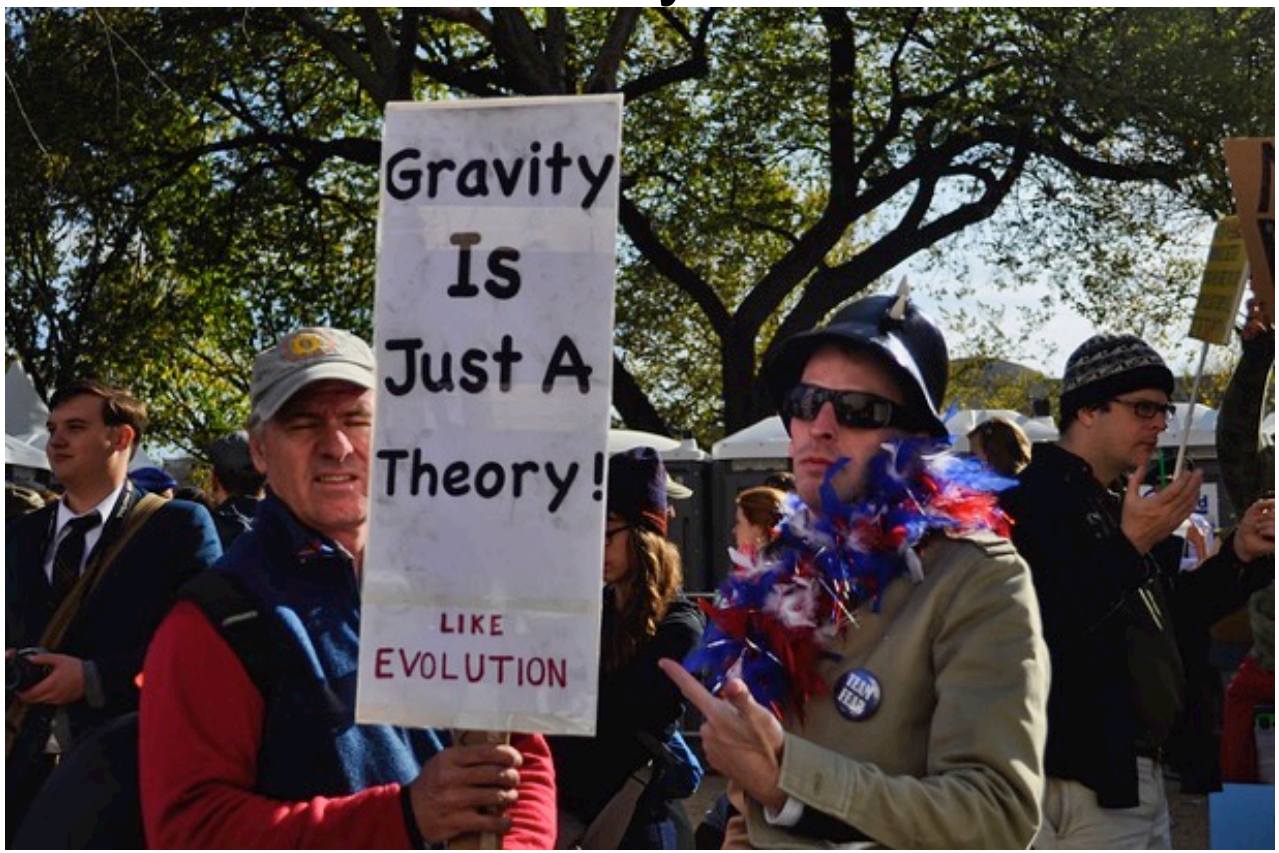
- Belief comes in degrees:

  "I fully believe that", "I tend to believe you", "This is hard to believe"

- A belief can be more or less justified
- If one believes that *p* is true with probability 0.7, is justified in believing this, and *p* is true, then one knows *p.*

Uncertainty of Knowledge

Uncertainty and limited degrees of confidence in a claim is not a reason to deny that it is knowledge



Uncertainty of Knowledge

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…but scientists should strive to achieve highest possible degree of confidence

Justification as a Social Enterprise

In science, individuals are requested to give reasons for their beliefs

- *Discussion with other scientists*



Justification as a Social Enterprise

In science, individuals are requested to give reasons for their beliefs

- *Discussion with other scientists*
- *Evaluation through other scientists*

Justification as a Social Enterprise

Reproducibility as a scientific virtue

- Other people should be able to assess the justification and come to the same conclusion on the same grounds
- for instance, reproduce an experiment and get the same result

Justification as a Social Enterprise

The peer review process in scientific journals

- Research papers are submitted to specialized scientific journals.
- Experts (1-5, often anonymous) read the research paper before recommending *accept, reject* or *revise.*
- The experts typically do not redo the experiment or study.
- They base their recommendations on how well the authors ***justify*** their claims (and whether the justification is original).

Justification as a Social Enterprise

The scientific ethos:

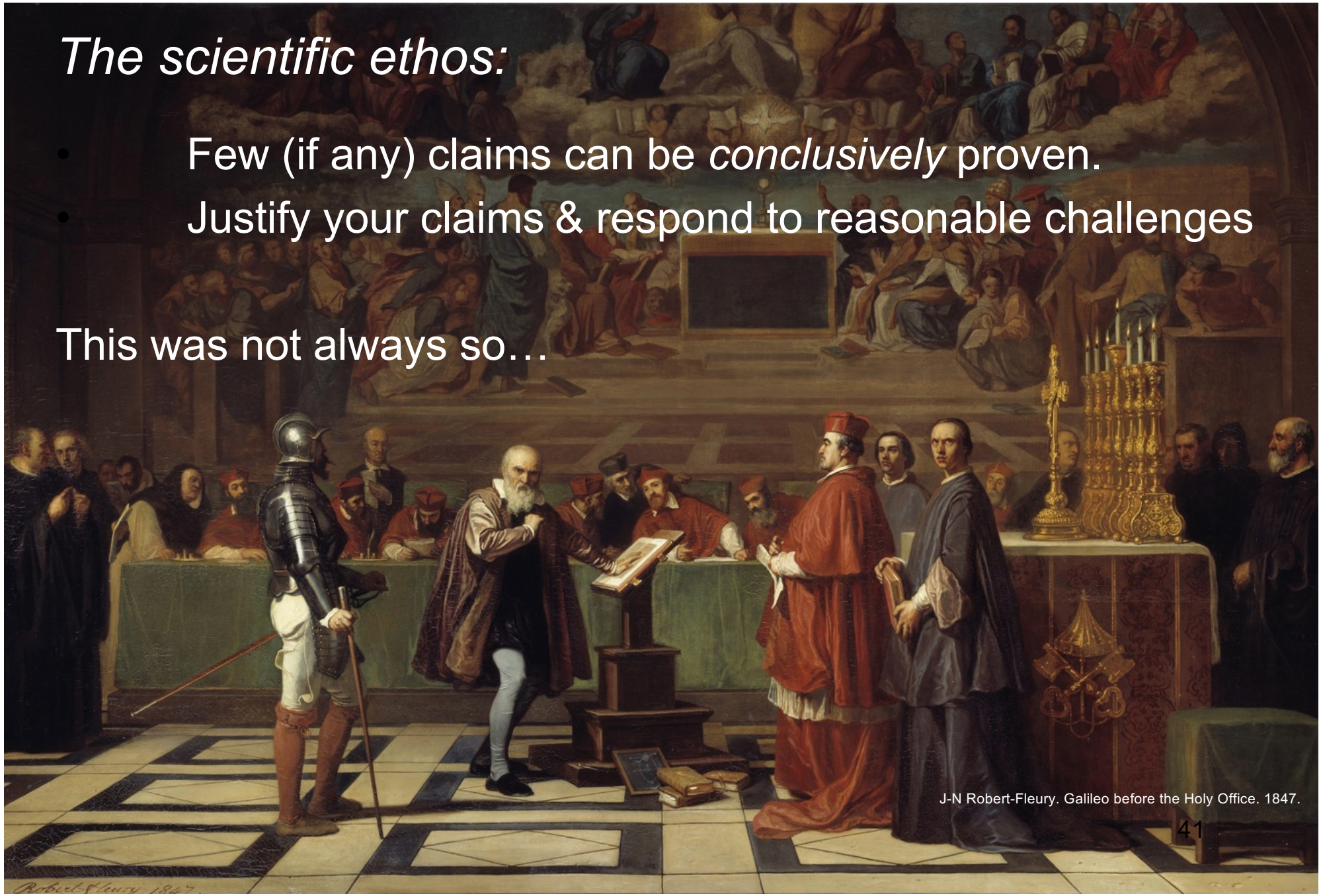
- Few (if any) claims can be *conclusively* proven.
- Justify your claims & respond to reasonable challenges to

Justification as a Social Enterprise

The scientific ethos:

- Few (if any) claims can be *conclusively* proven.
- Justify your claims & respond to reasonable challenges

This was not always so…



J-N Robert-Fleury. Galileo before the Holy Office. 1847.

Learning Results

- Distinguish methodology from methods
- Be able to discuss the presented arguments against scientificness and conventional justifications
- Explain the epistemically tool approach to method choice justification and its heuristic nature
- Discuss the major goals of science and how they relate to the concept of knowledge
- Explain the classical definition of knowledge
- Describe the concept of belief and the uncertainty of scientific knowledge
- Explain how scientific knowledge is justified, why epistemic diversity matters and how this relates to practices like public presentation, reproducibility and peer review